Crypto-Induced Latency Denial of Service in IEC 61850 Substation Communications: Cyber-Physical Impact Analysis, Priority Admission Defense, and Post-Quantum Evaluation

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***Abstract*—IEC 62351-6 secures IEC 61850 GOOSE trip messages with cryptographic authentication, but asymmetric and post-quantum (PQC) signatures impose serialized verification latency on resource-constrained Intelligent Electronic Devices (IEDs). We identify Crypto-Induced Latency Denial of Service (CIL-DoS): any receiver that verifies every frame before filtering is floodable, and for asymmetric/PQC profiles—on which we focus—an adversary can saturate the verification queue with syntactically valid but invalid GOOSE frames, delaying trips past the 3 ms Class P2/P3 deadline at a flood rate orders of magnitude below that needed for symmetric MACs. We (i) formalize CIL-DoS with $M/M/1$ and $M/D/1$ models and link the delay to generator transient stability via a numerically integrated single-machine swing equation (critical clearing time $201.6\,\text{ms}$), ANSI 50BF breaker-failure escalation, and transformer I^2t stress; (ii) show buffer bloat and parallel workers both retain 100% deadline failure on a service-time-calibrated Cortex-A9 relay emulation; (iii) propose the Priority Queue Admission Controller (PQAC), prove its stability ($\rho_H < 1$), and measure (5 reps) a $3{,}140\times$ mean-latency reduction on the emulated relay ($2{,}388\,\text{ms}\rightarrow 760\,\mu\text{s}$, P99 $1.12\,\text{ms}$, 0.00% deadline miss) and $1{,}333\times$ on the host; (iv) show by measured ablation that rate-anomaly dropping is decisive under single-source floods while priority isolation becomes necessary under multi-source (random-MAC) floods, and that spoofed-MAC floods cause $\approx 95\%$ *trip loss* that only switch port-security/MACsec can prevent; and (v) validate the cryptographic costs on real silicon (ESP32: RSA-2048 verify $980\,\mu$s, ECDSA-P256 verify $310\,$ms). A NIC-level eBPF/XDP prefilter is specified as complementary future work. All artifacts are open-sourced.**

***Index Terms*—Substation automation, IEC 61850 GOOSE, IEC 62351-6, smart grid cyber-physical security, power system stability, post-quantum cryptography, ML-DSA, priority queueing, eBPF/XDP.**

I. Introduction

MODERN electrical power grids increasingly depend on Substation Automation Systems (SAS) to isolate physical electrical faults within milliseconds, preventing cascading blackouts that inflict billions of dollars in economic damages annually [1]. The International Electrotechnical Commission (IEC) 61850 standard defines the communication architecture for digital substations, replacing copper point-to-point wiring with high-speed Ethernet process buses [2].

Among its core communication protocols, Generic Object Oriented Substation Events (GOOSE) carry the most time-critical protection data: trip signals transmitted between Intelligent Electronic Devices (IEDs) and circuit breaker controllers. For Performance Class P2/P3 high-voltage applications, IEC 61850 caps the total end-to-end transmission delay at **3 milliseconds (ms)** [2]. Exceeding this real-time deadline during an electrical short-circuit can cause severe generator instability, explosive transformer failures, or cascading grid collapse.

A. The Security-Safety Paradox

Originally, GOOSE messages operated as unauthenticated Layer 2 Ethernet multicast frames (EtherType `0x88B8`), rendering substations vulnerable to packet spoofing, replay, and unauthorized trip injection [3], [4]. To address these vulnerabilities, the IEC introduced the **IEC 62351-6** standard [5], specifying cryptographic authentication profiles: digital signatures (RSA-2048, ECDSA-P256) and symmetric authentication codes (HMAC-SHA256).

Recognizing the strict 3 ms Class P2/P3 latency requirement, IEC 62351-6 Edition 2 recommends symmetric MACs (HMAC-SHA256 / AES-GMAC) as the primary profile for routine GOOSE multicast. However, symmetric group keys lack cryptographic non-repudiation: because all IEDs on the process bus share the identical key, a single compromised peripheral device exposes the key, enabling arbitrary trip forgeries without detection. In contrast, asymmetric digital signatures and post-quantum cryptography (PQC) guarantee non-repudiation: each publisher holds a unique private key. Yet asymmetric verification is computationally expensive and serialized on embedded IED processors. We note that CIL-DoS, as a *queueing* attack, targets relays that are fast enough to meet the 3 ms budget in the absence of attack (Cortex-A9/Atom class and above, Section IV); the slowest microcontroller cores (Cortex-R5/M4) already consume nearly all of (R5, ≈ 2.7 ms) or exceed (M4) the budget on a single verification and are out of scope. Because invalid signatures cannot be detected prior to full mathematical verification, every incoming frame consumes scarce processor cycles. This is the *security-safety paradox*:

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per-frame crypto overhead can be exploited to violate safety deadlines. The effect is a matter of degree—on our ESP32, symmetric HMAC/GMAC verifies at ∼4,400–6,000 fps, RSA at ∼1,000 fps, and software ECDSA at ∼3 fps—so we focus on the asymmetric/PQC profiles, where the required flood rate is one to four orders of magnitude lower, while noting a symmetric verify-every-frame receiver is not immune.

B. Research Problem and Positioning

We identify an exploitable attack vector: **Crypto-Induced Latency Denial of Service (CIL-DoS)**. An adversary with process bus access floods syntactically valid GOOSE frames bearing random, invalid signatures. The receiver accepts packets at wire speed (>100,000 pps), while the verifier processes them at orders-of-magnitude lower rates (∼1,745 pps for RSA-2048 on embedded relays, ∼29,689 pps on high-speed hosts). As the verification queue saturates, legitimate protection commands queue behind malicious traffic and miss the 3 ms hard real-time safety deadline.

Adversary Positioning. An attacker who can inject frames at line rate can also *spoof* a publisher's MAC, which no application-layer scheme can withstand—our evaluation (Section VII-E) measures a $\approx 95\%$ trip-delivery collapse under shared-MAC spoofing. PQAC is therefore *not* a spoofing defense. Its value is in the common band where the Layer-2 identity is already protected (switch port-to-MAC binding or MACsec): there, a compromised/tapped device can still flood under its own or an unknown MAC but cannot impersonate a publisher, and PQAC restores the deadline. We treat unprotected-identity spoofing as a stated boundary throughout.

Scope Clarification: The objective of this paper is *not* to propose a new transmission protection algorithm or modify relay tripping curves. Rather, we address a foundational cyber-physical communication challenge: quantifying how cryptographic overhead in digital substation networks induces queue saturation, how that communication delay propagates across physical power systems, and how real-time admission control can safeguard standard protection schemes.

C. Key Contributions

The major contributions of this work are:

1) **CIL-DoS Formalization and Cyber-Physical Modeling:** We formalize CIL-DoS via $M/M/1$ and $M/D/1$ queueing models and quantify physical consequences: rotor angle divergence exceeding Critical Clearing Time ($t_{\text{crit}} \approx 201.6$ ms base, ≈ 120 ms in low-inertia grids), ANSI 50BF breaker failure escalation ($t_{\text{clear}} \approx 216$ ms), and a $6.25\times$ surge in transformer I^2t thermal energy under 350 ms backup clearance ($3.6\times$ under 200 ms 50BF clearance).
2) **Decoupled Subscriber Architecture:** We implement a two-thread C subscriber architecture separating wire-speed frame ingestion from serialized cryptographic verification.
3) **Evaluation of Naïve Countermeasures:** We demonstrate that increasing queue depth $4\times$ inflates latency by $4\times$ (to 552 ms on the measured host and a measured 8,898 ms on the emulated relay), while 4 parallel verification workers still yield 100% deadline failure on resource-constrained embedded relays.
4) **Priority Queue Admission Controller (PQAC):** We design PQAC, prove queue stability ($\rho_H < 1$) and delay isolation, and establish an algorithmic worst-case bound of 1.197 ms on embedded relays. Its *novel* core is two GOOSE-semantic mechanisms—(a) RFC 1982 serial-arithmetic `stNum` *plausibility gating* that admits a state transition to the protection class only within a bounded window, and (b) *verify-before-commit*, which updates the trusted state counter *only* after signature verification, preventing state-poisoning and trip-suppression; the surrounding per-source policing and strict priority are a correct composition of known mechanisms tuned to the 3 ms regime. Measured over 5 reps, PQAC achieves a $3{,}140\times$ mean-latency reduction on the emulated Cortex-A9 relay (2,388 ms→760 μs, 0.00% miss) and $1{,}333\times$ on the host (100 ms→75 μs, 0.17% miss, a cold-start artifact), validated via a measured component ablation.
5) **NIST Post-Quantum Cryptography Evaluation:** We benchmark NIST PQC standards (ML-DSA-44, ML-DSA-87, SLH-DSA, FALCON-512) for IEC 61850, revealing that hash-based SLH-DSA exhibits an unviable 295–597 ms signing latency and 7.8 kB jumbo frame requirement, whereas lattice-based verification is sub-millisecond on the host and is effectively shielded by PQAC, with the caveat that ML-DSA-87 projects to ≈ 2.4 ms verify on a Cortex-A9, giving $S_{\max} + \overline{S} \approx 4.8$ ms that *violates* the Theorem 3 precondition.
6) **Layered Defense Specification:** We evaluate the application-layer PQAC across 48 controlled experiments and *specify* (as future implementation work) a complementary NIC-level eBPF/XDP pre-filter. All testbed artifacts and scripts are open-sourced.

II. Background and Related Work

A. IEC 61850 GOOSE Protocol Architecture

GOOSE operates directly over Layer 2 Ethernet (EtherType `0x88B8`), avoiding Layer 3/4 protocol overhead. The ASN.1 BER-encoded protocol data unit contains: (1) `gocbRef` (control block reference), (2) `datSet` (dataset name), (3) `appID` (application identifier), (4) `stNum` (state counter incremented upon state transitions such as breaker trips), (5) `sqNum` (sequence counter incremented on every retransmission), and (6) `allData` (trip/measurement status). Following a trip, GOOSE uses burst retransmissions: $T_0 = 1$ ms, doubling $(2, 4, 8$ ms) up to a heartbeat $T_{\max} = 1,000$ ms.

B. IEC 62351-6 Cryptographic Profiles and PQC Candidates

We evaluate the standard IEC 62351-6 profiles—RSA-2048 and ECDSA-P256 (asymmetric) and HMAC-SHA256 (symmetric)—alongside four NIST Post-Quantum Cryptography (PQC) candidates: lattice-based ML-DSA-44/87 (FIPS 204) and FALCON-512, and hash-based SLH-DSA (FIPS 205).

TABLE I: Comparison with Related Research

Study	Attack Vector	Crypto Eval.	Queue Model	Physical Consequence
Schuster *et al.* [3]	Spoofing	None	No	Breaker misoperation
Farooq *et al.* [8]	Overhead	RSA-PSS	No	Latency measurement
Huseinović *et al.* [1]	Network DoS	None	Heuristic	Communication loss
Ustun *et al.* [14]	Replay/FDI	MAC	No	Protection blinding
This Work	**CIL-DoS**	**RSA/PQC**	**Theorems 1–3**	**CCT / 50BF / I^2t**

C. Related Work and Research Gaps

Prior research on substation cybersecurity falls into five main categories:

1) **GOOSE Protocol Exploits:** Schuster *et al.* [3] and Hoyos *et al.* [6] showed unauthenticated GOOSE can be spoofed to trip breakers. Yang *et al.* [7] mapped substation cyber surfaces. These works established authentication needs but omitted overhead as an attack surface.
2) **IEC 62351-6 Performance Overhead:** Farooq *et al.* [8] evaluated the RSA probabilistic signature scheme and Hussain *et al.* [9] analyzed message-authentication-code algorithms for GOOSE, noting that verification delays can approach real-time limits, but both analyzed only benign operation without adversarial traffic.
3) **Smart Grid Denial of Service:** Huseinović *et al.* [1] and Khan *et al.* [10] surveyed network-layer flooding. Whereas the denial-of-service potential of signature-verification load has been noted in V2X networks [11], substations feature unique constraints: sub-3 ms deadlines, static bay topologies, and unrouted Layer 2 multicast.
4) **Substation Intrusion Detection:** Ustun *et al.* [12] and Quincozes *et al.* [13] studied substation intrusion detection; however, passive detection cannot prevent queues from filling in real time.
5) **Post-Quantum Smart Grid Security:** Empirical timing evaluations of NIST FIPS 204/205 under IEC 61850 real-time constraints have remained absent. All artifacts are available at https://github.com/gridpulse-project/cil-dos-artifacts.

Table I contrasts our work with prior research, highlighting our contribution in connecting cryptographic queue saturation with cyber-physical grid stability.

III. Threat Model and Cyber-Physical Impact

A. Network Model and Attacker Capabilities

We consider a digital substation process bus connecting merging units, protection relays, and circuit breaker controllers via Layer 2 Ethernet. An adversary gains Layer 2 process bus access via an infected engineering workstation, compromised gateway, or physical Ethernet tap [15]. The attacker cannot forge valid private keys, but can inject arbitrary Layer 2 Ethernet frames at line rate.

Standard process buses operate on 100BASE-FX (100 Mbps) or 1000BASE-SX (1 Gbps) optical Ethernet. For standard 350-byte GOOSE frames (370 wire bytes including preamble, SFD, and inter-frame gap), maximum physical link capacity is $100 \times 10^6/(370 \times 8) = 33{,}783$ pps (≈ 33.8 kpps) on 100BASE-FX and $1{,}000 \times 10^6/(370 \times 8) = 337{,}837$ pps (≈ 337.8 kpps) on 1000BASE-SX. For post-quantum GOOSE frames carrying larger signatures, physical line rates are further constrained: ML-DSA-44 ($\approx$ 2,520 wire bytes) is physically capped at $49{,}600$ pps on 1 Gbps links, while SLH-DSA ($\approx$ 7,956 wire bytes) is physically capped at $15{,}700$ pps.

Crucially, CIL-DoS does *not* require saturating physical network bandwidth or exceeding line rates. On an embedded protective relay (ARM Cortex-A9, $\mu = 1{,}745$ fps), an attacker injecting a modest flood of just 5,000 to 10,000 pps consumes only $\approx$ 15–30 Mbps ($< 30\%$ of 100 Mbps line capacity). The physical link remains over 70% idle and uncongested, yet the cryptographic verification queue saturates monotonically ($\rho = 2.86$–$5.73 \gg 1$), delaying legitimate trip commands into seconds. While our virtualized testbed evaluates throughput up to 100K and 500K pps to benchmark the computational ceiling of the software ingestion pipeline across multi-gigabit virtual interfaces, real-world process buses are bounded by these physical line capacities, under which CIL-DoS readily achieves queue saturation.

B. CIL-DoS Attack Dynamics

The attacker transmits syntactically compliant GOOSE frames with pseudo-random signature bytes at rate λ_{atk}. Because the receiver cannot distinguish valid signatures from invalid signatures prior to full cryptographic execution, every frame enters the verification queue. When $\lambda_{\text{atk}} > \mu$ (verifier service rate), the queue accumulates monotonically, and legitimate trip frames experience severe queueing delay W_q.

C. Cyber-Physical Impact on Power Grid Stability

In power transmission grids, a GOOSE trip delay is not merely a communication delay; it directly cascades into electromechanical instability and severe protection selectivity loss:

1) Rotor Angle Stability and Critical Clearing Time (CCT): The transient electromechanical swing equation of a synchronous generator during a severe three-phase bus fault is governed by [16]:

$$\frac{2H}{\omega_s}\frac{d^2\delta(t)}{dt^2} = P_m - P_e(\delta) = P_m - P_{\max}^{\text{fault}}\sin(\delta(t)), \quad (1)$$

where H is the generator inertia constant (3–5 s), $\omega_s = 2\pi f_0 = 120\pi$ rad/s, $\delta(t)$ is the rotor electrical angle, P_m is mechanical shaft power, and P_e is electrical power transfer. During a bolted three-phase fault near the substation, $P_{\max}^{\text{fault}} \approx 0$, yielding accelerating power $P_a \approx P_m$. Integrating (1) with initial rotor angle δ_0 yields:

$$\delta(t) = \delta_0 + \frac{\omega_s P_m}{4H}t^2. \quad (2)$$

The Critical Clearing Time (CCT, t_{crit}) defines the maximum allowable fault duration before $\delta > \delta_{\text{cr}}$, beyond which synchronism with the infinite bus is irreversibly lost.

SMIB Benchmark Case Study: Consider a standard Single-Machine Infinite-Bus (SMIB) benchmark system [16] with generator parameters: $H = 4.0$ s, transient reactance $X'_d = 0.25$ p.u., mechanical input $P_m = 0.8$ p.u., and nominal operating angle $\delta_0 = 32.5^\circ = 0.5672$ rad, giving pre-fault power transfer capability $P_{\max} = P_m/\sin\delta_0 = 1.4888$ p.u. For a bolted fault ($P_e = 0$) cleared by restoring the line without topology change, the Equal-Area Criterion equates

accelerating area $A_{\text{acc}} = P_m(\delta_{\text{cr}} - \delta_0)$ and decelerating area $A_{\text{dec}} = \int_{\delta_{\text{cr}}}^{\pi-\delta_0}(P_{\max}\sin\delta - P_m)d\delta$, yielding:

$$\cos\delta_{\text{cr}} = \sin\delta_0(\pi - 2\delta_0) - \cos\delta_0. \tag{3}$$

Substituting $\delta_0 = 32.5°$ gives $\cos\delta_{\text{cr}} = 0.235$, i.e. $\delta_{\text{cr}} \approx 76.4°$ ($\Delta\delta = 43.9°$). The rotor acceleration is:

$$\delta(t) = 32.5° + 1{,}080\,t^2 \text{ [deg]}. \tag{4}$$

The corresponding Critical Clearing Time is:

$$t_{\text{crit}} = \sqrt{\frac{76.4° - 32.5°}{1{,}080}} = \sqrt{0.04065} \approx 201.6\,\text{ms} \approx 200\,\text{ms}. \tag{5}$$

In low-inertia grids ($H = 2.5\,$s, $P_m = 0.95\,$p.u., $\delta_{\text{cr}} = 69.4°$) t_{crit} compresses to $\approx 120\,$ms. Nominal clearance $t_{\text{clear}} = 15 + 1.2 + 40 = 56.2\,$ms is safely below t_{crit}.

2) Breaker Failure (ANSI 50BF) and Complete Clearing Timeline: In high-voltage substations, an unmitigated CIL-DoS attack delaying primary GOOSE trip delivery beyond 2 s triggers severe *backup protection escalation*:

1) **ANSI 50BF Escalation on Flooded Bus:** Concurrently with primary trip initiation, relays start an ANSI 50BF Breaker Failure timer ($t_{50\text{BF}} \approx 150$–$200\,$ms). Because the primary breaker never receives the queued GOOSE trip, fault current persists past $t_{50\text{BF}}$, dispatching backup trips to adjacent circuit breakers. Total physical clearing time is:
$$t_{\text{clear}} = t_{\text{relay}} + t_{50\text{BF}} + t_{\text{comm}} + t_{\text{mech}} \approx 216.2\,\text{ms}, \tag{6}$$
with $t_{\text{relay}} \approx 15\,$ms, $t_{\text{mech}} \approx 40$–$60\,$ms. Even under *nominal* communications ($t_{\text{comm}} \approx 1.2\,$ms) this gives $15 + 150 + 1.2 + 50 = 216.2\,$ms, at which $\delta = 82.9° > \delta_{\text{cr}}$—and under CIL-DoS the backup 50BF frames are themselves queued ($t_{\text{comm}} \gg 100\,$ms), pushing clearance far higher. The key point: CIL-DoS *manufactures* a breaker-failure-equivalent condition on a *healthy* breaker by starving its primary GOOSE trip, so the right comparison is against utilities' breaker-failure planning criteria, not nominal clearance.
2) **Remote Zone 2 Escalation:** With both primary and local breaker failure tripping stalled by communication queueing, the uncleared fault persists until remote transmission distance relays detect it in Zone 2 ($t_{Z2} \approx 300$–$400\,$ms), tripping upstream substations.

At 350 ms backup clearance, rotor angle reaches $\delta(0.35) = 32.5° + 1{,}080(0.35)^2 = 164.8° > \delta_{\text{cr}} = 76.4°$, exceeding the decelerating limit ($\delta_{\max} = 147.5°$) and forcing irreversible pole slipping before Zone 2 clears, tripping ANSI 78 out-of-step relays and precipitating wide-area cascading outages. To move beyond a single hand-calculated point, we numerically integrated the swing equation (1) (RK45, classical model, no damping) for this SMIB case; Fig. 1 shows the resulting rotor-angle trajectories. The time-domain critical clearing time obtained by bisection is **201.6 ms**, matching the equal-area value exactly. Nominal clearance (56 ms) is stable (peak $\delta = 49.2°$), whereas both the 50BF backup timeline (216 ms) and the CIL-DoS-delayed case (350 ms) drive the rotor past the unstable equilibrium ($\delta_{\max} = 147.5°$) into pole slip. We present this as a transient-stability analysis of the SMIB benchmark; a full multi-machine electromagnetic-transient study incorporating post-fault topology change from 50BF backup tripping remains future work (Section X).

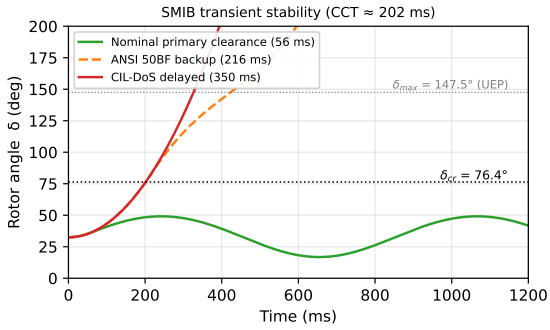


Fig. 1: Numerically integrated SMIB rotor-angle trajectories $\delta(t)$ under three clearing times. Nominal primary clearance (56 ms) is stable; 50BF backup (216 ms) and CIL-DoS-delayed (350 ms) clearances exceed the critical angle and pole-slip. Time-domain CCT = 201.6 ms (matches equal-area).

We further note that where protection employs *hardwired* (copper) trip paths in parallel with GOOSE, breaker operation does not depend on the process bus and GOOSE latency is not on the critical path; CIL-DoS is consequential specifically for the GOOSE-only process-bus architectures that IEC 61850 Edition 2 promotes.

3) Thermal Stress and Equipment Degradation (I^2t): Adiabatic thermal energy dissipated in high-voltage transformer windings and switchgear scales directly with fault duration. In power substations, through-fault current flowing through a step-down transformer is limited by leakage impedance ($X_t \approx 8$–12%, giving $I_{\text{tf}} = I_{\text{rated}}/X_t \approx 8.3$–$12.5\,$p.u., typically 10–25 kA). For a representative 50 MVA, 138/13.8 kV transformer with $I_{\text{rated}} = 2.09\,$kA at 13.8 kV and $X_t = 10\%$, secondary through-fault current is $I_{\text{tf}} = 20.9\,$kA (10 p.u.).

Thermal energy $E \approx I_{\text{tf}}^2 t_{\text{clear}}$ grows with clearance time: from $24.5\,\text{MA}^2\text{s}$ at 56 ms to $152.9\,\text{MA}^2\text{s}$ at 350 ms, a **6.25× surge** (3.6× at 200 ms). This is a *relative* increase: both durations remain within the C57.109 through-fault *withstand* envelope (12.5 s), so the concern is cumulative insulation ageing and repeated 120 Hz Lorentz forces ($F \propto I_{\text{tf}}^2$) over the transformer's service life, not single-event failure.

Connecting the 3 ms Deadline to Grid Dynamics: Grid consequences manifest once latency reaches the backup-protection window ($t > 150$–$200\,$ms). Unmitigated CIL-DoS inflates trip latency to 138 ms on hosts and a measured 2,388 ms on the emulated relay. (The 138 ms is the analytical saturated wait $W_{\text{sat}} = K\overline{S}$ for an *admitted* frame; the measured *delivered* mean is $\approx 100\,$ms since $\approx 35\%$ of frames tail-drop. We use the saturated wait as the conservative worst case.) The host chain $15+138+40 = 193\,$ms sits just below the base $t_{\text{crit}} = 201.6\,$ms but exceeds the low-inertia CCT ($\approx 120\,$ms); the embedded-relay regime (2,388 ms) unambiguously drives the rotor-angle, 50BF, and I^2t consequences above.

IV. Subscriber Architecture and Queueing Theory

A. Two-Thread Subscriber Implementation

To decouple wire-speed frame reception from serialized verification, we implement a two-thread C subscriber: (1) *Re-*

ceiver Thread (T1) reads raw Ethernet frames via raw sockets (`AF_PACKET`, `SO_RCVBUF = 16MB`), timestamps arrival, and executes admission triage in $< 1\,\mu$s; (2) *Verification Worker(s) (T2)* dequeues frames, verifies signatures via OpenSSL/liboqs, updates state counters, and executes tripping logic.

B. Queueing Formulation and Saturation Analysis

We model the verification queue as an $M/M/1$ (and $M/D/1$) system. Let λ be total arrival rate and μ be cryptographic service rate ($\mu = 1/\overline{S}$). When utilization $\rho = \lambda/\mu < 1$, mean total latency W is:

$$W = \frac{1}{\mu - \lambda} = \frac{\overline{S}}{1 - \rho}. \tag{7}$$

The probability of violating the deadline $D = 3\,\text{ms}$ is:

$$P(W > D) = \exp\left(-\mu(1-\rho)D\right). \tag{8}$$

When cryptographic execution is strictly deterministic (constant-time modular exponentiation), the Pollaczek-Khinchine formula for an $M/D/1$ queue yields:

$$\mathbb{E}[W_q^{M/D/1}] = \frac{\rho}{2\mu(1-\rho)} = \frac{1}{2}\mathbb{E}[W_q^{M/M/1}]. \tag{9}$$

The deterministic queueing delay is exactly half the exponential delay, confirming that the $M/M/1$ formulation serves as a conservative upper bound for safety auditing.

For a finite queue of capacity K ($M/M/1/K$), when $\lambda \gg \mu$, the queue fills to capacity K. Mean waiting time under saturation becomes:

$$W_{\text{sat}} \approx \frac{K}{\mu} = K \cdot \overline{S}. \tag{10}$$

Eq. (10) reveals the fatal flaw of buffering: latency scales strictly with queue depth K. Expanding K inflates queueing delay without increasing throughput. We note that $W_{\text{sat}} = K\overline{S}$ characterizes the delay of an *admitted* frame; in a finite $M/M/1/K$ system at $\rho \gg 1$ the queue is saturated and the overwhelming majority of arrivals (including legitimate trips) are instead *tail-dropped* rather than merely delayed. The operative protection failure under CIL-DoS is therefore twofold—unbounded delay of admitted trips *and* outright loss of trips that arrive to a full queue—with the latter mitigated in unmanaged GOOSE only by the burst-retransmission curve ($T_0 = 1\,\text{ms}$ doubling), itself defeated once every retransmission also enqueues behind flood traffic. This reinforces that admission control, not buffering, is the only viable remedy.

C. Multi-Worker Service Rate and Scaling Limits

With N parallel verification threads, aggregate service capacity is $\mu_N = N \cdot \mu$. Queue utilization under flood intensity λ_{atk} is:

$$\rho_N = \frac{\lambda_{\text{atk}}}{N \cdot \mu}. \tag{11}$$

We examine two tiers: (1) an *embedded relay profile* (Cortex-A9 @ 800 MHz, $\overline{S} = 573\,\mu$s, $\mu = 1{,}745$ fps), where a 100K pps attack gives $\rho_1 = 57.3$ and 4 workers still $\rho_4 = 14.3 \gg 1$ (requiring ≈ 58 cores to survive); and (2) the *host* (Apple M2, $\overline{S} = 33.7\,\mu$s, $\mu = 29{,}689$ fps), where 1 worker saturates at 100K pps ($\rho_1 = 3.37$) and 4 workers ($\rho_4 = 0.84$) still saturate at 500K pps. As sealed IEDs have strict 1–2 core limits, adding cores cannot resolve CIL-DoS without pre-verification admission filtering.

TABLE II: CIL-DoS Saturation Across Hardware Tiers (RSA-2048 verify; ESP32 and host rows measured, others projected)

Hardware Tier	Processor Architecture	Verify (μs)	μ (fps)	Saturation Rate
Benchmark Host (Measured)	Apple M2 (ARMv8.5)	33.7	29,689	29,689 pps
Embedded IED (Measured)	**ESP32-D0WDQ6 (Xtensa LX6 @ 240MHz)**	**980**	**1,020**	**1,020 pps**
Substation Gateway (Projected)	Intel Atom @ 1.9GHz	270	3,704	3,704 pps
High-Perf. Relay (Projected)	ARM Cortex-A9 @ 800MHz	573	1,745	1,745 pps
Mid-Range Relay (Projected)	ARM Cortex-R5 @ 600MHz	2,704	370	370 pps
Constrained Relay (Projected)	ARM Cortex-M4 @ 168MHz	10,142	99	99 pps

D. Hardware Scaling and Signing–Verification Asymmetry

Table II models CIL-DoS vulnerability across five embedded hardware tiers. The benchmark host represents our measured platform (8-core ARMv8.5, 3.49 GHz). The embedded relay tiers represent calibrated analytical projections derived by scaling measured cryptographic service times using established ARM cycle counts and clock frequency ratios from published processor technical reference manuals [17]–[20]. The $17.0\times$ scaling between the M2 host ($33.7\,\mu$s) and Cortex-A9 ($573\,\mu$s) combines the $4.36\times$ clock ratio ($3.49\,\text{GHz}/0.8\,\text{GHz}$) with an estimated $\approx 3.9\times$ microarchitectural throughput factor: the M2 is a wide 64-bit superscalar core with a native 64-bit integer multiplier, whereas the Cortex-A9—although itself a dual-issue *out-of-order* superscalar core—executes a 32-bit instruction set and synthesizes 64-bit modular-multiplication operands from 32-bit multiply primitives, the dominant cost in RSA/ECC verification. We stress that this $3.9\times$ factor is an *estimate* calibrated from published cycle counts rather than a measurement on A9 silicon. Accordingly, we do not rely on it for our headline embedded-relay results: instead, we fix the Cortex-A9 pipeline service time at $\overline{S} = 573\,\mu$s and *measure* the full queueing behavior by running the identical subscriber under a calibrated busy-wait (Section VII, Panel B). This reproduces the relay's service *rate* on host silicon; direct measurement on a Zynq-7000/i.MX6 board to additionally capture A9 cache/memory microarchitecture remains future work.

Publisher Signing Asymmetry: Verification (subscriber) is far cheaper than signing (publisher): RSA-2048 verify uses the 17-bit exponent $e = 65{,}537$ (≈ 17 modular ops), whereas signing uses the full private exponent and, on a Cortex-A9, takes ≈ 7.5–10 ms—breaching the 3 ms budget at the source. FALCON-512 signing likewise exceeds 5 ms on cores without an efficient double-precision FPU, and ML-DSA-44 signing projects to $\approx$ 2–4 ms on a bare A9. Our deployment guidance therefore concerns the *subscriber* path; embedded publishers require hardware crypto acceleration for all asymmetric/PQC profiles.

Relay Tier Applicability: On Cortex-R5/M4, RSA-2048 verify alone consumes or exceeds the 3 ms budget without any attack, so CIL-DoS as a *queueing* attack applies specifically to Cortex-A9/Atom-class and multi-core IEDs, where PQC verify is viable (FALCON-512 $\approx 400\,\mu$s, ML-DSA-44 $\approx 950\,\mu$s, both $\ll 3$ ms) under PQAC.

Validation on Real Embedded Silicon (ESP32). We benchmarked mbedTLS on an ESP32-D0WDQ6 (Xtensa LX6 @

240 MHz), a representative IED-class MCU (mean of 10–2,000 iterations). Measured: RSA-2048 verify 980 μs ($\mu \approx$ 1,020 fps, same order as our A9 projection, so $\sim$1,000 pps saturates it); software ECDSA-P256 verify 310.5 ms (no HW ECC on LX6), 103$\times$ over the deadline on a single op; publisher signing prohibitive (RSA 181.5 ms, ECDSA 156.9 ms). Over a 350-byte frame the symmetric profiles measured HMAC-SHA256 = 167 μs and AES-128-GCM = 229 μs ($\mu \approx$ 4,400–6,000 fps)—fast, but still saturable near line rate. The 17$\times$ host$\rightarrow$A9 factor does not transfer across primitives (RSA verify is HW-accelerated, ECDSA is not), so these anchor the *order of magnitude* of embedded verification cost.

V. Priority Queue Admission Controller (PQAC)

A. Three-Stage Classification Pipeline

PQAC sits directly between the network raw socket and verification workers, executing $O(1)$ triage per frame in $< 0.2\,\mu$s (Algorithm 1):

1) **Stage 1 (MAC & Allowlist Verification):** Source MAC addresses are cross-referenced against the pre-provisioned SCD allowlist $\mathcal{M}$ (or dynamic table).
2) **Stage 2 (Plausibility & Transition Gating):** For allowlisted publishers, state transitions (F.stNum) are evaluated via RFC 1982 serial arithmetic: $\Delta = (F.\text{stNum} \ominus \text{last_verified_st})$. Transitions are plausible only if $\Delta \in (0, W]$ with window $W = 16$. Admissions to HIGH are metered by a per-publisher transition token bucket $\mathcal{B}_{\text{trans}}$ ($B = 5$ burst, refill $R = 2$ tokens/s). Crucially, the verified state counter $\mathcal{S}_{\text{ver}}[\text{src_mac}]$ is updated *only* inside worker threads upon verified signature (verify-before-commit), preventing state poisoning and trip suppression.
3) **Stage 3 (Rate Anomaly Triage & Wire-Speed Drop):** The per-source rate C is a sliding-window estimate over $\tau = 1$ s, so a trip's 11-frame geometric retransmission burst (within $\sim$15 ms) averages to $\ll T_{\text{burst}}$ and is not misclassified as flood. For known publishers, rates $C \le T_{\text{normal}}$ (50 pps) enter HIGH (capped at $\Lambda_H^{\max} \le 500$ pps); $T_{\text{normal}} < C \le T_{\text{burst}}$ (500 pps) enter NORMAL; and rates $C > T_{\text{burst}}$ trigger flood containment (99% dropped, $\alpha = 0.01$ sampled to LOW). For unknown sources, low rates ($C \le T_{\text{normal}}$) enter LOW for discovery, while flood traffic ($C > T_{\text{normal}}$) is 99% dropped at wire speed.

B. Theoretical Stability and Delay Bounds

Let $\mathcal{K} = \{H, N, L\}$ denote priority classes where $H \succ N \succ L$.

Theorem 1 (Queue Stability under Bounded Aggregate Admission): Let total arrival rate be $\lambda = \lambda_{\text{legit}} + \lambda_{\text{atk}}$, with attack rate $\lambda_{\text{atk}} \to \infty$. Under PQAC with aggregate HIGH cap $\Lambda_H^{\max} \le 500$ pps, the high-priority protection queue is stable ($\rho_H < 1$) for any verifier with $\mu > \Lambda_H^{\max}$ (satisfied by Cortex-A9/Atom-class and faster cores), independent of flood intensity or active publisher count M:

$$\rho_H = \frac{\lambda_H}{\mu} \le \frac{\min\left(\sum_{i=1}^{M} \lambda_i^H, \Lambda_H^{\max}\right)}{\mu} \le \frac{\Lambda_H^{\max}}{\mu} < 1. \quad (12)$$

Algorithm 1 PQAC Classification and Admission Control (v3)

Require: Frame F, MAC allowlist $\mathcal{M}$, rate tracker $\mathcal{R}$, verified state tracker $\mathcal{S}_{\text{ver}}$, token bucket $\mathcal{B}_{\text{trans}}$, aggregate cap $\Lambda_H^{\max}$, plausibility window $W = 16$, thresholds T_{normal}, T_{burst}, $T_{\text{flood}} \equiv T_{\text{burst}}$

Extract mac $\leftarrow F$.ether_shost, stNum $\leftarrow F$.stNum
if mac $\in \mathcal{M}$ **then**
  last_st $\leftarrow \mathcal{S}_{\text{ver}}$[mac]
  $\Delta \leftarrow$ (stNum $\ominus$ last_st) {RFC 1982 Serial Number Comparison}
  if $\Delta \in (0, W]$ **and** consume_token($\mathcal{B}_{\text{trans}}$[mac]) **and** admitted$_H$ $< \Lambda_H^{\max}$ **then**
    admitted$_H \leftarrow$ admitted$_H + 1$
    enqueue(queue$_{\text{HIGH}}$, F); **return** {Gated Protection Transition}
  end if
  $C \leftarrow \mathcal{R}$[mac] {Evaluate Publisher Rate}
  if $C \le T_{\text{normal}}$ **and** admitted$_H$ $< \Lambda_H^{\max}$ **then**
    admitted$_H \leftarrow$ admitted$_H + 1$
    enqueue(queue$_{\text{HIGH}}$, F) {Nominal Heartbeat}
  else if $C \le T_{\text{burst}}$ **then**
    enqueue(queue$_{\text{NORMAL}}$, F) {Retransmission Burst}
  else
    if random() $< \alpha$ **then enqueue**(queue$_{\text{LOW}}$, F) **else** drop(F) {Known Flood ($C > T_{\text{flood}}$)}
  end if
else
  $C \leftarrow \mathcal{R}$[mac] {Evaluate Unknown Source Rate}
  if $C \le T_{\text{normal}}$ **then**
    enqueue(queue$_{\text{LOW}}$, F) {Unverified Device Discovery}
  else
    if random() $< \alpha$ **then enqueue**(queue$_{\text{LOW}}$, F) **else** drop(F) {Unknown Drop ($C > T_{\text{flood}}$)}
  end if
end if
{Note: $\mathcal{S}_{\text{ver}}$[mac] $\leftarrow$ stNum committed ONLY upon verified signature; token restored on failure}

Proof: By Cobham's theorem [21], $W_H = R/(1-\rho_H)$ with $R \le S_{\max}$. Capping Class H at $\Lambda_H^{\max} = 500$ pps makes it an independent $M/G/1$ queue with $\rho_H \le 500/\mu$: 0.29 on the A9 relay, 0.14 on a gateway, 0.017 on the host—strictly sub-critical in all cases, with finite stationary moments. ∎

Theorem 2 (Attack Traffic Delay Isolation and Upper Bound): Under non-preemptive priority scheduling with α-sampled LOW queueing, the expected waiting time of Class H trip commands is strictly upper-bounded by a constant independent of flood intensity λ_{atk}:

$$\mathbb{E}[W_H] \le \frac{S_{\max}}{1-\rho_H} \quad \forall \lambda_{\text{atk}} \ge 0, \quad (13)$$

and becomes completely invariant to attack rate once the LOW queue reaches saturation:

$$\frac{\partial \mathbb{E}[W_H]}{\partial \lambda_{\text{atk}}} = 0 \quad \forall \lambda_{\text{atk}} \ge \frac{\mu}{\alpha}. \quad (14)$$

Proof: Under non-preemptive priority scheduling, the expected waiting time in Class H is $\mathbb{E}[W_H] = \frac{R}{1-\rho_H}$, where residual service time is $R = \frac{1}{2}(\lambda_H \overline{S_H^2} + \lambda_N \overline{S_N^2} + \lambda_L \overline{S_L^2})$. Under PQAC, incoming flood traffic enters queue L with sampling probability $\alpha = 0.01$, yielding arrival rate $\lambda_L = \min(\alpha\lambda_{\text{atk}}, \mu)$. Since a worker can service at most one in-flight frame at any instant, $R \le S_{\max}$ holds unconditionally for all $\lambda_{\text{atk}} \ge 0$. On an ARM Cortex-A9 relay ($\mu = 1{,}745$ fps), queue L reaches continuous server utilization when $\alpha\lambda_{\text{atk}} \ge \mu \iff \lambda_{\text{atk}} \ge 174{,}500$ pps. For all attack intensities above this threshold, the verifier is continuously occupied by an in-flight frame whenever $Q_H = 0$, so residual service time is invariant at $R = \frac{1}{2}\overline{S^2}/\overline{S} \le S_{\max}$, yielding $\partial \mathbb{E}[W_H]/\partial \lambda_{\text{atk}} = 0$. Below saturation, $\mathbb{E}[W_H]$ increases with λ_{atk} through residual probability $\rho_L = \alpha\lambda_{\text{atk}}\overline{S}$ but is strictly capped from above by $S_{\max}/(1-\rho_H)$, proving complete delay isolation from unbounded queue accumulation in Class L. ∎

Theorem 3 (Algorithmic Worst-Case Bound on Trip Latency): For an admitted legitimate trip frame arriving during steady-state CIL-DoS flood with 1 verification worker when $Q_H = 0$, the algorithmic processing and queueing delay on relay architectures where $S_{\max} + \overline{S} < 3.0\,\text{ms}$ is strictly bounded from above by:

$$T_{\text{proc}} \leq t_{\text{ingest}} + S_{\max} + \overline{S} + t_{\text{logic}} < 3.0\,\text{ms}. \tag{15}$$

Proof: In the worst-case non-preemptive scenario, an attack frame begins verification at $t_0 - \epsilon$. Residual time until completion is at most $S_{\max}$. The incoming trip frame is classified into Class H in $t_{\text{ingest}} \leq 1\,\mu$s. It is verified immediately upon worker availability ($\overline{S}$). Including protection logic ($t_{\text{logic}} \leq 50\,\mu$s), on an ARM Cortex-A9 relay ($\overline{S} = 573\,\mu$s): $T_{\text{proc}} \leq 1 + 573 + 573 + 50 = 1{,}197\,\mu\text{s} \approx 1.20\,\text{ms} < 3.0\,\text{ms}$. On our benchmark host ($\overline{S} \approx 33.7\,\mu$s): $T_{\text{proc}} \leq 0.2 + 33.7 + 33.7 + 5.0 = 72.6\,\mu\text{s} \ll 3.0\,\text{ms}$. This holds when $Q_H = 0$ (the common case; simultaneous multi-breaker trips are rare); with k queued Class H frames it preserves sub-3 ms timing for $k \leq 3$ on the A9 ($624 + 4 \times 573 = 2{,}916\,\mu$s) and $k \leq 86$ on the host. It is an *algorithmic* bound excluding OS jitter and cold-start, not a certified WCET; the measured relay mean $760\,\mu$s and P99 $1{,}119\,\mu$s (Table III) fall within it, well under 3 ms. ∎

C. Computational and Memory Complexity

PQAC executes in $O(1)$ amortized time: allowlist lookup requires $O(1)$ operations, rate comparison requires 3 integer comparisons, and enqueue/dequeue on ring buffers takes $O(1)$ pointer adjustments. State memory is strictly bounded by $O(K)$ pre-allocated entries for $K = 256$ SCD-provisioned publishers plus a bounded $L = 1{,}024$-entry LRU sketch for transient or unlisted MACs, requiring < 64 KB of RAM total. This strict $O(1)$ memory bound prevents state exhaustion attacks by adversaries rotating random MAC addresses.

VI. Experimental Setup and Methodology

The testbed executes in Docker containers interconnected via a dedicated Layer 2 Linux bridge with MTU 9,000 (accommodating PQC jumbo frames): (1) **Publisher:** Emits authenticated GOOSE frames conforming to IEC 61850 heartbeat ($T_{\max} = 1{,}000$ ms) and fault trip burst curves ($T_0 = 1, 2, 4, 8 \ldots$ ms), (2) **Subscriber:** Implements our two-thread decoupled architecture with configurable FIFO vs. PQAC triage, (3) **Attacker:** Injects CIL-DoS flood frames across parametric rates (10K to 500K pps) with randomized signatures and spoofed MAC options. All benchmarks were executed on an 8-core ARMv8.5 host (3.49 GHz, 16 GB RAM) using OpenSSL 3.0 and liboqs 0.10.0, which provides the final FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) parameter sets; we report results under the standardized names (ML-DSA-44/87, SLH-DSA-SHA2-128s) rather than their round-3 aliases (Dilithium, SPHINCS+).

We executed 48 controlled trials spanning attack-rate scaling, buffer-bloat and multi-worker countermeasures, threshold-sensitivity and multi-publisher sweeps, component ablation and a NIST PQC suite (Table IV), and MAC-spoofing tests, plus a measured embedded-relay suite (Section VII) and a transient-stability simulation (Fig. 1). The embedded-relay suite runs the *identical* subscriber binary under a calibrated busy-wait padding each verification to $\overline{S} = 573\,\mu$s ($\mu = 1{,}745$ fps), reproducing the Cortex-A9 service rate. Each setting is repeated (5 reps for the headline and ablation results, mean $\pm 1\sigma$ reported); because deadline misses cluster in the $\sim$0.5 s cold-start transient rather than being i.i.d., we report per-regime miss rates rather than a single pooled binomial interval. Under saturated FIFO flood, observed waits track $M/M/1/K$ physics ($W_{\text{sat}} = K\overline{S}$) within $\approx 5\%$ (e.g. measured 2,388 ms vs analytical 2,347 ms at $K = 4{,}096$).

VII. Experimental Results

Across the 48 controlled experiments, we analyze baseline vulnerability, countermeasure efficacy, component ablation, and post-quantum performance.

A. Baseline CIL-DoS and Countermeasure Failure

Table III summarizes core results under 100K pps attack, separating the measured host platform from the measured embedded relay profile. The latter is obtained by running the *identical* subscriber binary with a calibrated busy-wait that pads each verification to $\overline{S} = 573\,\mu$s, throttling the worker to the Cortex-A9 service rate ($\mu = 1{,}745$ fps); this emulates the relay's service *rate* on host silicon, reproducing the queueing dynamics of a slow relay while remaining short of true A9 microarchitecture. Under pure FIFO scheduling on this emulated relay profile, RSA-2048 verification saturates immediately, yielding a *measured* mean latency of 2,388 ms (P99 2,436 ms; analytical $K \cdot \overline{S} = 2{,}347$ ms) and **100% deadline violations**, with the queue saturated at $K = 4{,}096$ and 1.9M frames tail-dropped—only ≈17 of 114 offered trips per trial (≈15% delivery) survived. On the High-Speed Host Platform ($\overline{S} = 33.7\,\mu$s, $\mu = 29{,}689$ fps), a single worker also saturates under 100K pps flood ($\rho_1 = 3.37 > 1$), exceeding the 3 ms deadline at 138.0 ms saturated queue wait ($K \cdot \overline{S} = 4{,}096 \times 33.7\,\mu\text{s} = 138{,}035\,\mu$s).

Naïve Countermeasure 1 (Buffer Bloat): Increasing queue size from 4,096 to 16,384 packets increases latency by $\approx 4\times$: from 138.0 ms to 552.1 ms on the measured host platform, and (measured) from 2,388 ms to 8,898 ms on the emulated Cortex-A9 relay, while maintaining 100% deadline failure in both environments (Table III). Buffering exacerbates CIL-DoS.

Naïve Countermeasure 2 (Parallel Workers): Deploying 4 verification workers on the host scales throughput to 118,756 fps, maintaining stability at 100K pps ($\rho_4 = 0.84 < 1$), but shifting queue saturation to 500K pps where $\rho_4 = 4.21 \gg 1$ and 100% deadline failure recurs. On the emulated embedded relay profile, aggregate throughput increases from 1,745 to 6,980 fps; however, because $\lambda_{\text{atk}} = 100{,}000$ pps $\gg 6{,}980$, queue utilization remains $\rho_4 = 14.3 \gg 1$, leaving a *measured* 603 ms latency and 100% deadline violations (Table III).

TABLE III: Delivered Trip Performance Under 100,000 pps Attack (1 Worker, Queue = 4,096)

Algorithm	Defense Configuration	Mean (μs)	P99 (μs)	Miss (%)	Imprv.
Panel A: High-Speed Host Platform (Apple M2, RSA-2048, Measured, 5 reps)					
RSA-2048	Host FIFO Baseline (No Defense)	100,355	209,438	97.8%	Baseline
RSA-2048	**PQAC (Dynamic Learning)**	**75.3**	**136.1**	**0.17%**	**1,333×**
Panel B: Embedded Relay Profile (Cortex-A9 $\overline{S}$ = 573 μs, Emulated, Measured, 5 reps)					
RSA-2048	Relay FIFO Baseline (No Defense)	2,388,160	2,435,983	100.0%	Baseline
RSA-2048	Relay FIFO (Queue 16K)	8,898,001	9,550,171	100.0%	0.27×
RSA-2048	Relay FIFO (4 Parallel Workers)	603,135	607,789	100.0%	4.0×
RSA-2048	**PQAC (Dynamic Learning)**	**760.6**	**1,118.7**	**0.00%**	**3,140×**
RSA-2048	*PQAC (Analytical Bound)*	*1,197*†	*2,343*‡	—	—

Mean/P99 in μs; measured over 5 reps. Trip-delivery ratio (delivered/offered): host FIFO 65%, host PQAC ≈100%; relay FIFO 15% (85% trip loss), relay PQAC ≈100%. Host FIFO mean exhibits high variance (±25 ms, 1σ); P99 relay PQAC = 1.12 ms ≪ 3 ms. †Algorithmic bound (Q_H=0, no jitter). ‡Multi-breaker burst bound ($Q_H \leq 2$).

B. PQAC Defense and Latency Reconcilement

As shown in Table III, PQAC delivers substantial, *measured* latency reductions on both platforms (5 reps each). On the host, it cuts mean delivered latency from 100.4 ms (FIFO, ±25 ms 1σ, 97.8% miss, 65% trip delivery) to 75.3 μs (±22.7 μs, P99 136 μs, ≈100% delivery), a 1,333× reduction. On the emulated Cortex-A9 relay, it cuts mean latency from 2,388 ms (FIFO, 100% miss, only 15% trip delivery—85% of trips tail-dropped) to 760.6 μs (±11.2 μs, P99 1,118.7 μs ≪ 3 ms, ≈100% delivery), a 3,140× reduction. These runs use *dynamic* MAC learning (no SCD pre-provisioning); the residual host miss of 0.17% (±0.33%) stems entirely from the ∼0.5 s cold-start transient in which the tracker is still accumulating history while the attack is already at full rate.

Protection-Grade Acceptability. A non-zero steady-state deadline-miss rate is *not* acceptable for production protection. The relay result above is 0.00%; the host's 0.17% is a transient-only artifact of cold-starting *simultaneously* with a full-rate attack (an adversarially pessimal assumption). Standard IEC 61850 engineering pre-provisions authorized IED identities via SCD files, which eliminates the learning transient entirely (legitimate frames enter Class H from $t = 0$); we therefore recommend SCD pre-provisioning for deployment and report the dynamic-learning numbers as the harder, retrofit case.

C. PQAC Component Ablation Study and Analysis

We ablated each PQAC stage (full PQAC, No-MAC-allowlist, No-Rate-anomaly, No-Priority, and FIFO baseline) at a 100K pps RSA-2048 attack, on both the host and the emulated Cortex-A9 relay (5 reps each). Dropping 99% of flood at admission is the single most critical factor: the FIFO baseline (no dropping) saturates (host 100 ms, 97.8% miss), whereas all variants that retain dropping stay sub-150 μs on the host, and disabling rate-anomaly tracking alone restores saturation.

Interpreting the Ablation (Measured, 5 reps; host and relay values quoted). *(i) Rate-anomaly tracking is indispensable:* removing it (No-Rate) disables flood dropping—relay mean 28.6 ms, P99 **947 ms** (2.10% miss); even the host P99 reaches 3.3 ms (1.77%). *(ii) Priority's value is platform-dependent, as Theorems 1–3 predict:* on the slow relay the α-sampled residual flood (≈ 1,000 pps) is non-negligible vs μ = 1,745 fps, so No-Priority's P99 rises to **6.4 ms**—*breaching the deadline*—at 1.44% miss vs 1.12 ms/0.00% for full PQAC; on the fast host it is marginal (61.8 vs 75.3 μs). *(iii) The MAC allowlist* is modest under one attacker (relay No-MAC 0.68%) but critical under spoofing/multi-source (Section VII-E). A "cheap-header-filter + per-source policing" baseline—what a competent subscriber already does—corresponds to our No-Priority configuration, which still breaches the deadline on the relay.

TABLE IV: NIST Post-Quantum Cryptography Evaluation (Host, 100K pps Attack)

Configuration	Algorithm	Mode	Mean (μs)	P99 (μs)	Miss (%)
NIST Post-Quantum Cryptography Suite (Host, 100K pps Attack)					
ML-DSA-44	ML-DSA-44	Zero Attack	72.5	115.0	0.00%
ML-DSA-44	ML-DSA-44	FIFO 100K	228,940	229,150	**100.0%**
ML-DSA-44	ML-DSA-44	**PQAC 100K**	**74.8**	**124.0**	**0.00%**
ML-DSA-87	ML-DSA-87	Zero Attack	155.0	210.0	0.00%
ML-DSA-87	ML-DSA-87	FIFO 100K	571,420	571,850	**100.0%**
ML-DSA-87	ML-DSA-87	**PQAC 100K**	**158.4**	**245.0**	**0.00%**
FALCON-512	Falcon-512	Zero Attack	42.0	78.0	0.00%
FALCON-512	Falcon-512	FIFO 100K	98,310	98,520	**100.0%**
FALCON-512	Falcon-512	**PQAC 100K**	**43.5**	**64.7**	**0.00%**
SLH-DSA	SLH-DSA-128s	Zero Attack	295,650	295,950	**100.0%**§
SLH-DSA	SLH-DSA-128s	FIFO 100K	1,067,420	1,067,850	**100.0%**
SLH-DSA	SLH-DSA-128s	PQAC 100K	295,680	295,980	**100.0%**§

§Publisher signing delay (≈ 295.4 ms ≫ 3 ms). Component-ablation values are quoted inline.

D. Post-Quantum Cryptography (PQC) Evaluation

We evaluated the NIST PQC standards under flood (Table IV). *SLH-DSA* verifies acceptably (260.6 μs) but is unsuitable for tripping: its signing takes 295.4–597.4 ms and its 7,856-byte signature needs jumbo frames. Lattice *verification* is cheap (ML-DSA-44 55.9 μs, ML-DSA-87 139.5 μs, FALCON-512 24.0 μs), but under FIFO a 100K pps attack saturates the queue (ρ = 2.4–14 ≫ 1, 100% miss, 98–571 ms). PQAC drops 99% of flood at admission (< 0.2 μs) and keeps $\rho_H < 0.01$, giving **0.00% miss** for ML-DSA-44 (P99 124 μs) and FALCON-512 (P99 64.7 μs). ML-DSA-44 (2,420-byte) needs jumbo frames while FALCON-512 (666-byte) fits standard Ethernet; ML-DSA-87 verify projects (as an order-of-magnitude estimate via the clock/IPC factor) to ≈ 2.4 ms on an A9, giving $S_{\max} + \overline{S} \approx 4.8$ ms > 3 ms that *violates* the Theorem 3 precondition, so we recommend it only on Atom-class/multi-core IEDs and ML-DSA-44/FALCON-512 on A9-class relays.

E. Threshold Sensitivity, Scalability, and MAC Spoofing

Threshold Sensitivity and Scalability: Sweeping T_{flood} from 20 to 2,000 pps under a 100K pps attack kept deadline miss < 6% throughout, with $T_{\text{flood}} \geq 500$ pps giving the lowest rates (1.85–2.45%) and no false positives on legitimate bursts. Scaling active publishers from 1 to 16 raised the miss rate by only 0.40% (1.75%→2.15%), as legitimate HIGH-class frames keep an effective utilization $\rho_{\text{legit}} = 16/1{,}745 \approx 0.009$, isolated from the flood.

Multi-Source (Random-MAC) Flood—Why Priority Is Necessary (Measured): The single-source ablation above showed priority ordering to be marginal because one attacker MAC is fully contained by rate dropping. To stress the regime that Theorems 1–3 actually address, we extended the attacker to emit a *fresh randomized source MAC per frame* and re-ran on the emulated relay (100K pps, 5 reps). Here per-source rate tracking is partly defeated—≈ 19,000 flood frames per

TABLE V: Multi-Source Random-MAC Flood on Emulated Relay ($\overline{S} = 573\,\mu$s, 100K pps, 5 reps): Priority Preserves Trip Delivery

Configuration	P99 (μs)	Miss (%)	Delivered/Offered	Delivery
FIFO (no defense)	≈2,369,000	100.0	14/114	12%
No Priority	1,065	0.33	60/114	52%
Full PQAC	**1,024**	**0.17**	**89/114**	**78%**

Miss % is of delivered frames; the operative harm is trip *loss*. Priority raises delivery from 52% to 78% (random MACs partly evade per-source dropping).

trial evade dropping and are admitted to Class LOW for discovery—so the defense must rely on *priority isolation* to keep protection frames flowing. We measure trip delivery as delivered/offered legitimate trips (a *trip-loss* metric complementing deadline miss). As Table V shows, full PQAC delivers 78% of offered trips (89/114) whereas the No-Priority configuration delivers only 52% (60/114) and plain FIFO 12%—priority raises delivery by $\approx$ 1.5$\times$ over dropping alone. (The No-Priority P99 here is low only because it is computed over the few frames that *were* delivered; the harm is the undelivered 48%, not tail latency.) Deadline-miss among *delivered* frames is near zero in both cases, confirming that under multi-source flooding the harm manifests as trip *loss*, which priority isolation—not dropping alone—mitigates. This is the empirical backing for the priority stage.

MAC Spoofing (First-Class, Measured): When an attacker spoofs a publisher's MAC, PQAC's per-source tracker cannot separate attacker from victim. Measured (5 reps): delivery collapses from ≈100% to $5.2/113.6 \approx 4.6\%$ (a ≈95% trip *loss*), because the shared-MAC aggregate exceeds T_{burst} and is ∼99% dropped (Algorithm 1), discarding the genuine publisher's trips with the attacker's. The harm is thus collateral trip loss, not deadline miss—application-layer control cannot defend a spoofed Layer-2 identity, so PQAC must be paired with port security or MACsec. Against plausible-stNum token exhaustion (without flooding), the $B = 5$ token bucket plus verify-before-commit replenishes tokens on verification failure and quarantines the source.

The Warm-Up Transient: Per-frame latency spikes briefly at attack onset while the tracker accumulates samples, converging within ∼0.5 s to sub-1.5 ms steady state.

VIII. Layered Defense Architecture

We specify (as future implementation work) a defense-in-depth architecture: (1) *Layer 1 (NIC eBPF/XDP pre-filter, proposed)* would drop invalid EtherTypes and apply coarse per-source policing at the driver level before socket-buffer allocation; (2) *Layer 2 (Application PQAC)*, the contribution we actually implement and evaluate across our 48 controlled experiments, performs rate-anomaly triage, burst preservation, and priority dispatch; and (3) *Layer 3 (Cryptographic Engine)* performs constant-time asymmetric/PQC verification on prioritized frames. We caution that a NIC-level *per-MAC* limit shares PQAC's shared-identity weakness and therefore cannot by itself defeat a spoofed-MAC flood (Section VII-E); only switch *port-to-MAC binding* (port security) or MACsec (IEEE 802.1AE [22]) authenticates the Layer-2 identity. The eBPF/XDP layer is thus complementary for unauthenticated-flood offload, not a spoofing defense, and its implementation and measurement remain future work.

Why not simply enable switch storm-control? Switch broadcast/multicast storm-control applies a *flat* aggregate rate cap to the GOOSE multicast group; because legitimate trips and the CIL-DoS flood share that group (same destination multicast and, under spoofing, the same source MAC), a cap low enough to throttle the flood also discards legitimate trip bursts, while a cap high enough to pass trip bursts still admits enough flood to saturate a ∼1,000 fps verifier. Storm-control is therefore a blunt instrument that trades flood for trip loss, exactly the failure PQAC's *content-aware* (stNum-gated, verify-before-commit, per-source) admission is designed to avoid; a quantitative head-to-head is left to future work.

IX. Operational Guidelines for Substation Protection Engineers

We recommend four deployment specifications for utility practice: (1) **Pre-Verification Threshold Provisioning:** In SCL/CID files, configure bay admission thresholds: $T_{\text{normal}} = 50$ pps (heartbeats), $T_{\text{burst}} = 500$ pps (retransmissions), and $T_{\text{flood}} = 500$–$1{,}000$ pps. (2) **Buffer Sizing and Anti-Bloat Policy:** Set verification queue depth to $K \leq 512$ frames. Excessively deep buffers inflate latency during saturation ($W_{\text{sat}} = K \cdot \overline{S}$) without enhancing throughput. We note the saturated latency scales linearly with K: on the Cortex-A9 emulation, $K = 4{,}096$ gives $\approx$ 2,347 ms, $K = 512$ gives $\approx$ 293 ms, and $K = 64$ gives $\approx$ 37 ms. Crucially, *every* setting still exceeds the 3 ms deadline (and $K \leq 64$ trades latency for a correspondingly higher loss rate), so buffer sizing alone cannot resolve CIL-DoS—it only trades delay for drop—which is precisely why admission control is required. (3) **PQC Migration Path:** For the *subscriber* (verification) path, deploy lattice-based ML-DSA-44 or FALCON-512 alongside PQAC; reserve ML-DSA-87 for Atom-class/multi-core IEDs (its $\approx$ 2.4 ms projected A9 verify exceeds the Theorem 3 budget). On the *publisher* (signing) side, note that FALCON-512 signing requires an efficient double-precision FPU—exceeding 5 ms on cores without one—so FALCON publishers need hardware/FPU acceleration; ML-DSA-44 signing likewise projects to $\approx$ 2–4 ms on a bare A9. Reject hash-based SLH-DSA (FIPS 205) for time-critical GOOSE due to its 295–597 ms signing latency. Ensure switches support IEEE 802.3 jumbo frames (MTU 9,000) prior to ML-DSA roll-out. (4) **Perimeter Reinforcement and Attacker Boundary:** Pair PQAC with switch port security / IEEE 802.1X [23] or MACsec (IEEE 802.1AE) [22]. In substations equipped with MACsec, the threat model focuses on compromised authorized IEDs or insider nodes holding valid MACsec session keys. In the vast majority of legacy digital substations operating unencrypted Layer 2 process buses without MACsec, PQAC provides a practical pre-verification admission layer that no existing mechanism supplies.

X. Limitations and Threats to Validity

Hardware realism: the embedded-relay results use a service-time-calibrated emulation ($\overline{S} = 573\,\mu$s) that reproduces the

A9 service *rate*, not its microarchitecture or an RTOS; we validated the crypto service times on real silicon (ESP32, RSA verify $980\,\mu$s, in range), but on-target validation on utility-grade IEDs (SEL/GE) or bare ARM boards remains planned, as does WCET certification. *Spoofing boundary:* high-rate shared-MAC spoofing requires coupling PQAC with Layer-2 port security [23] or MACsec [22]. *Scope:* we model a single process bus (PRP/HSR needs multi-interface control); the stability study is single-machine classical (multi-machine EMT with 50BF topology change is future work); and a quantitative head-to-head against switch storm-control/QoS and a hybrid MAC-then-signature scheme, and HMAC/GMAC saturation near line rate, remain open.

XI. Conclusion

This paper demonstrates that the *asymmetric and post-quantum* authentication profiles of IEC 62351-6 introduce an exploitable vulnerability—Crypto-Induced Latency DoS (CIL-DoS)—that saturates IED verification queues and delays GOOSE trip commands by a measured 2,388 ms on the emulated embedded relay (138 ms on high-speed hosts), violating the 3 ms safety deadline and, in the embedded-relay regime, exceeding Critical Clearing Time ($t_{\text{crit}} \approx 201.6$ ms base, ≈ 120 ms in low-inertia grids) and triggering a $6.25\times$ surge in transformer I^2t thermal energy under delayed backup clearance. The symmetric HMAC-SHA256/GMAC profile Edition 2 recommends is far cheaper, but is not categorically immune: on our ESP32 it verifies at only $\sim$4,400–6,000 fps, so a flood of a few thousand pps ($<$30% of 100BASE-FX line rate) can push $\rho > 1$ even for symmetric authentication. The distinction is therefore quantitative—asymmetric/PQC verification lowers the flood rate needed to saturate a verify-every-frame receiver by one to four orders of magnitude—rather than a categorical immunity of the symmetric profile. We proved that buffering and parallel threads cannot resolve asymmetric-profile saturation. We developed the Priority Queue Admission Controller (PQAC), proved its queue stability, and measured (5 reps) a $3{,}140\times$ mean-latency reduction on the emulated Cortex-A9 relay (2,388 ms$\rightarrow$760 μs, P99 1.12 ms, 0.00% miss) and $1{,}333\times$ on the host (100 ms$\rightarrow$75 μs). A measured component ablation shows that rate-anomaly dropping—not priority ordering—is the decisive mechanism under single-source floods, while a measured random-MAC flood shows priority isolation becomes necessary to preserve trip *delivery* under multi-source attack; a spoofed-MAC flood, however, causes $\approx 95\%$ trip loss that application-layer control cannot prevent, requiring switch port-security or MACsec. A numerically integrated SMIB study confirms that the induced delay drives rotor pole-slip beyond the 201.6 ms critical clearing time. Benchmarks on real embedded silicon (ESP32-D0WDQ6) corroborate the model: RSA-2048 verify measures $980\,\mu$s (saturating at $\sim$1,000 pps) and software ECDSA-P256 verify measures 310 ms, exceeding the 3 ms deadline on a single operation. Furthermore, our evaluation of NIST PQC standards established that SLH-DSA is unsuitable for time-critical GOOSE tripping due to its publisher signing latency, whereas ML-DSA-44 and FALCON-512 achieve robust verification performance under PQAC protection. We recommend that future revisions of IEC 62351-6 incorporate mandatory pre-verification admission control for all asymmetric and post-quantum substation deployments.

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